

Initiating Search

November 4, 2025, 9:37 PM

• Search:

CAS Lexicon Search:

Eschweiler-Clarke methylation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (129)	 References	View Results
Exported: Retrieved Related Reaction Results (90)	 Reactions	View Results

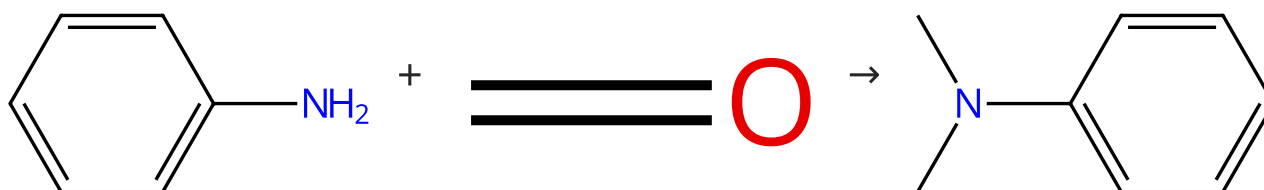
Copyright © 2025 American Chemical Society (ACS). All Rights Reserved.

Internal use only. Redistribution is subject to the terms of your CAS SciFinder License Agreement and CAS information Use Policies.

 Reactions (50)[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (113) Suppliers (188) Suppliers (90)

31-313-CAS-15461166

Steps: 1 Yield: 100%

A solvent-free and formalin-free Eschweiler-Clarke methylation of amines

1.1 Reagents: Ethanedioic acid hydrate (1:2)

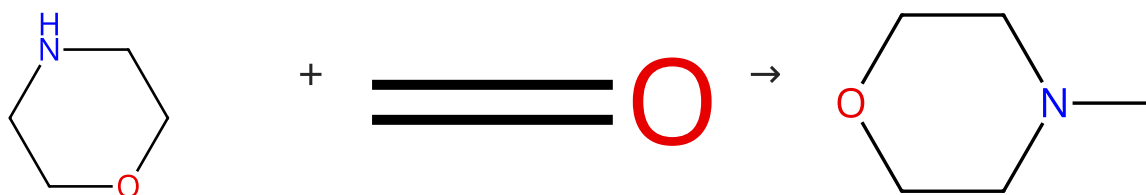
By: Rosenau, Thomas; et al

Experimental Protocols

Synthetic Communications (2002), 32(3), 457-465.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (88) Suppliers (188) Suppliers (81)

31-313-CAS-13060274

Steps: 1 Yield: 100%

A solvent-free and formalin-free Eschweiler-Clarke methylation of amines

1.1 Reagents: Ethanedioic acid hydrate (1:2)

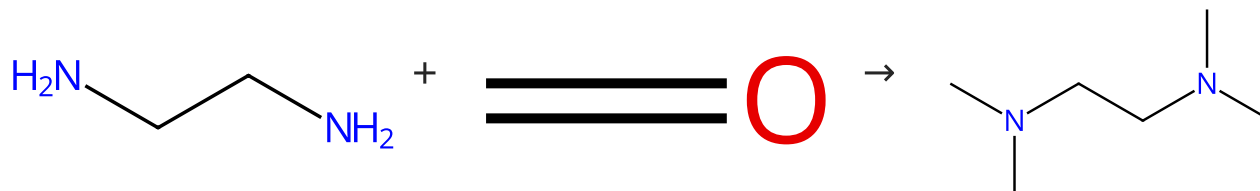
By: Rosenau, Thomas; et al

Experimental Protocols

Synthetic Communications (2002), 32(3), 457-465.

Scheme 3 (2 Reactions)

Steps: 1 Yield: 80-96%

 Suppliers (86) Suppliers (188) Suppliers (117)

31-313-CAS-2067007

Steps: 1 Yield: 96%

A solvent-free and formalin-free Eschweiler-Clarke methylation of amines

1.1 Reagents: Ethanedioic acid hydrate (1:2)

By: Rosenau, Thomas; et al

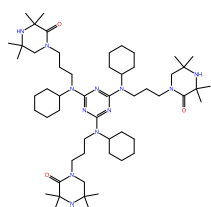
Experimental Protocols

Synthetic Communications (2002), 32(3), 457-465.

31-313-CAS-13175565	Steps: 1 Yield: 80%	Copper(I) halide complexes with N,N'-diallyl-N,N',N'-tetramethylethylenediaminium (L^{2+}). Synthesis and crystal structures of the complexes $[L_{0.5}CuCl_2]$, $[L_{0.5}CuCl_{0.72}Br_{1.28}]$, and $[L_{0.5}CuBr_2]$
1.1 Reagents: Formic acid Solvents: Water; 11 h, reflux		By: Monchak, M. M.; et al
1.2 Reagents: Hydrochloric acid Solvents: Water		Russian Journal of Coordination Chemistry (2011), 37(2), 143-148.
1.3 Reagents: Potassium hydroxide Solvents: Water		
Experimental Protocols		

Scheme 4 (1 Reaction)

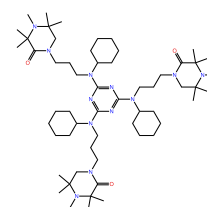
Steps: 1 Yield: 95%



+



→



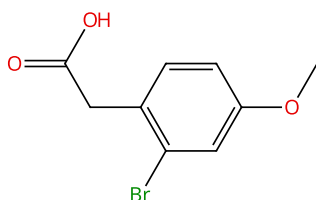
Supplier (1)

Suppliers (188)

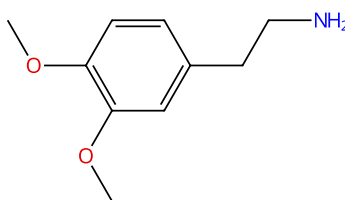
31-313-CAS-6462618	Steps: 1 Yield: 95%	Process for methylating a hindered piperazine nitrogen atom in an inert nonaqueous solvent
1.1 Reagents: Formic acid, Water Solvents: Toluene		Assignee: Goodrich, B. F., Co. European Patent Organization, EP429996 A2 1991-06-05

Scheme 5 (1 Reaction)

Steps: 1 Yield: 94%



+



+

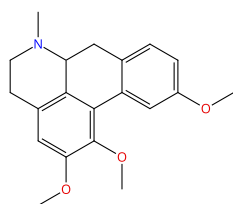


→

Suppliers (81)

Suppliers (64)

Suppliers (188)

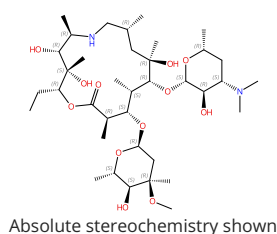
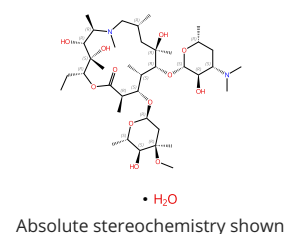
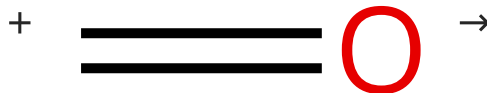


Suppliers (4)

31-614-CAS-37550607 Steps: 1 Yield: 94% 1.1 Reagents: 1-Hydroxybenzotriazole Solvents: Dichloromethane; 30 min, 0 °C 1.2 Reagents: Triethylamine, 1-Ethyl-3-(3'-dimethylaminopropyl) carbodiimide hydrochloride; 6 h, rt 1.3 Reagents: Phosphorus pentachloride Solvents: Dichloromethane; 0 °C; 18 h, rt 1.4 Reagents: Sodium borohydride Solvents: Methanol; 30 min, 0 °C; 2 min, 0 °C → rt 1.5 Reagents: Water 1.6 Reagents: Sodium borohydride Solvents: Methanol; 30 min, 0 °C; 8 h, 0 °C → rt 1.7 Reagents: Hydrochloric acid Solvents: Methanol, Water; 7 h	Green preparation of (chiral)aporphine alkaloid compounds Assignee: Zhijian Biotechnology (Shanghai) Co., Ltd. China, CN116675642 A 2023-09-01 PatentPak available
--	---

Scheme 6 (1 Reaction)

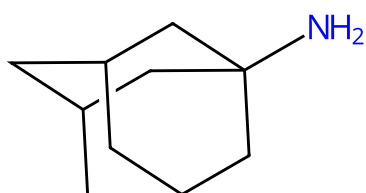
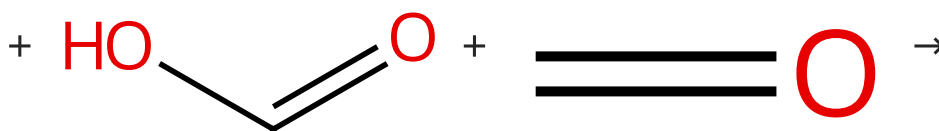
Steps: 1 Yield: 94%

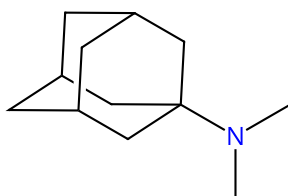

 Suppliers (61)

 Suppliers (14)

31-313-CAS-15003089 Steps: 1 Yield: 94% 1.1 Solvents: Formic acid; rt → reflux; 6 h, reflux 1.2 Reagents: Ethanol 1.3 Reagents: Sulfuric acid Solvents: Water; pH 3.8	Synthesis of azithromycin By: Ma, Min; et al Jingxi Huagong (2006), 23(8), 781-783.
---	--

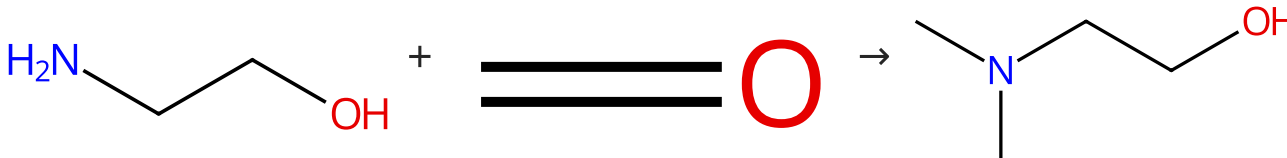
Scheme 7 (1 Reaction)

Steps: 1 Yield: 93%

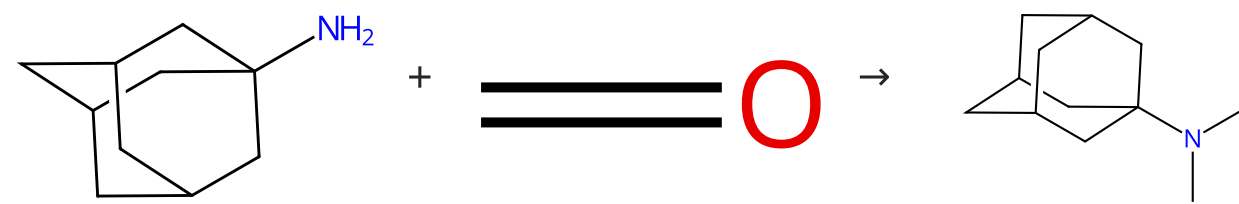

 Suppliers (139)

 Suppliers (176)

 Suppliers (188)

 Suppliers (25)

31-313-CAS-12905166 Steps: 1 Yield: 93% 1.1 90 °C; 18 h, 98 °C 1.2 Reagents: Sodium hydroxide Solvents: Water; pH 12 - 13	Synthesis of N,N,N-trimethyl-1-adamantyl quaternary ammonium hydroxide and technology development By: Hu, Jian; et al Advanced Materials Research (Durnten-Zurich, Switzerland) (2012), 391-392(Pt. 2), 1159-1163.
--	---

Scheme 8 (1 Reaction)   Suppliers (127)  Suppliers (188)  Suppliers (78)	Steps: 1 Yield: 93%
---	---

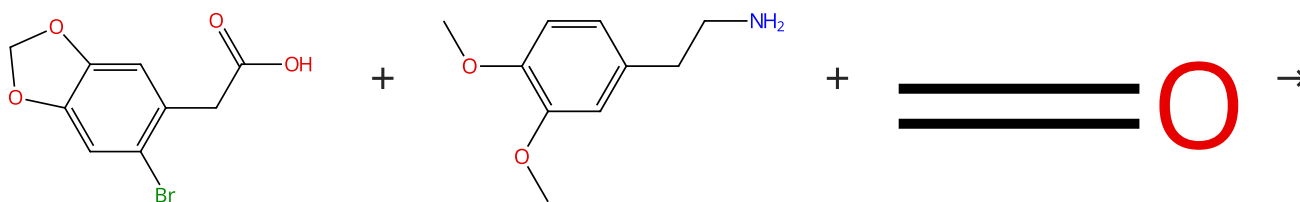
31-313-CAS-6328773 Steps: 1 Yield: 93% 1.1 Reagents: Ethanedioic acid hydrate (1:2) Experimental Protocols	A solvent-free and formalin-free Eschweiler-Clarke methylation of amines By: Rosenau, Thomas; et al Synthetic Communications (2002), 32(3), 457-465.
---	---

Scheme 9 (1 Reaction)   Suppliers (139)  Suppliers (188)  Suppliers (25)	Steps: 1 Yield: 93%
---	---

31-313-CAS-5552762 Steps: 1 Yield: 93% 1.1 Reagents: Formic acid Solvents: Water; 18 h, 98 °C 1.2 Reagents: Sodium hydroxide Solvents: Water; pH 12 - 13	Synthesis of N,N,N-trimethyl-1-adamantyl quaternary ammonium salt By: Hu, Jian; et al Huagong Xinxing Cailiao (2011), 39(8sp), 92-93, 122.
--	---

Scheme 10 (1 Reaction)

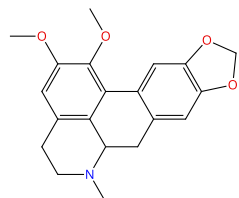
Steps: 1 Yield: 92%



Suppliers (38)

Suppliers (64)

Suppliers (188)



Suppliers (2)

31-614-CAS-37550608

Steps: 1 Yield: 92%

Green preparation of (chiral)aporphine alkaloid compounds

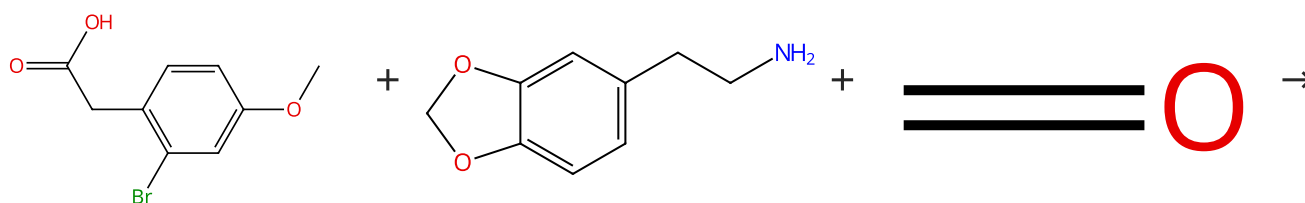
- 1.1 **Reagents:** 1-Hydroxybenzotriazole
Solvents: Dichloromethane; 30 min, 0 °C
- 1.2 **Reagents:** Triethylamine, 1-Ethyl-3-(3'-dimethylaminopropyl) carbodiimide hydrochloride; 6 h, rt
- 1.3 **Reagents:** Phosphorus pentachloride
Solvents: Dichloromethane; 0 °C; 18 h, rt
- 1.4 **Reagents:** Sodium borohydride
Solvents: Methanol; 30 min, 0 °C; 2 min, 0 °C → rt
- 1.5 **Reagents:** Water
- 1.6 **Reagents:** Sodium borohydride
Solvents: Methanol; 30 min, 0 °C; 8 h, 0 °C → rt
- 1.7 **Reagents:** Hydrochloric acid
Solvents: Methanol, Water; 7 h

Assignee: Zhijian Biotechnology (Shanghai) Co., Ltd.
China, CN116675642 A 2023-09-01

[PatentPak](#) available

Scheme 11 (1 Reaction)

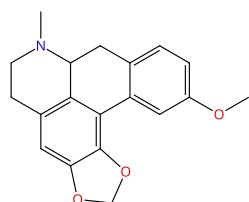
Steps: 1 Yield: 92%



Suppliers (81)

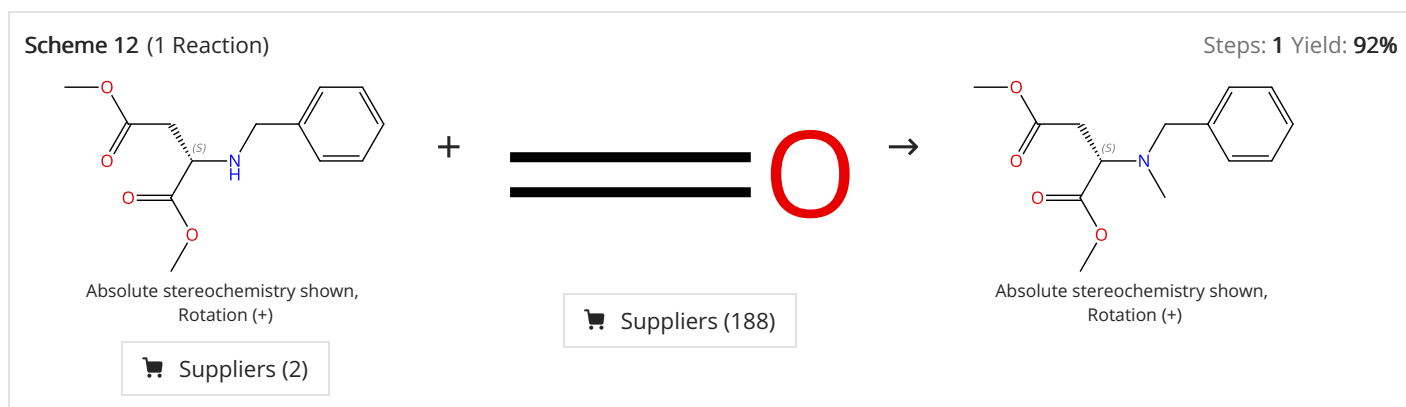
Suppliers (39)

Suppliers (188)

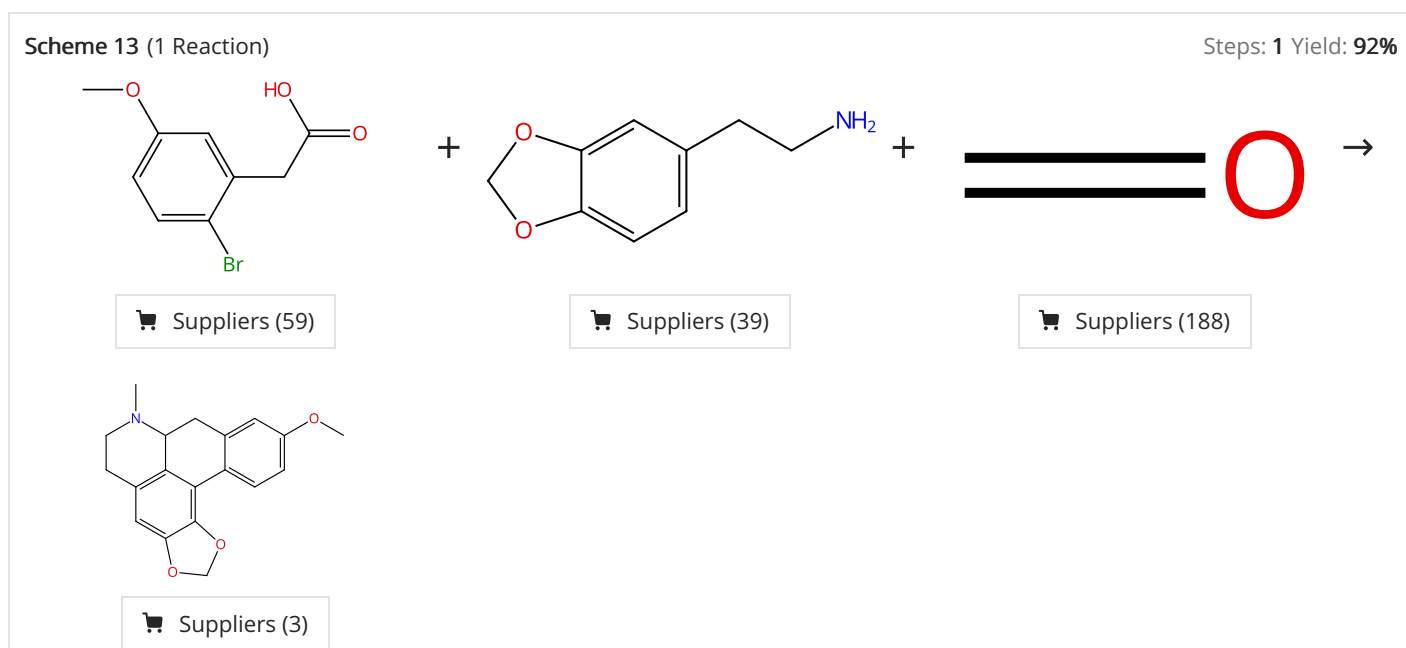


Suppliers (7)

31-614-CAS-37550614 Steps: 1 Yield: 92% 1.1 Reagents: 1-Hydroxybenzotriazole Solvents: Dichloromethane; 30 min, 0 °C 1.2 Reagents: Triethylamine, 1-Ethyl-3-(3'-dimethylaminopropyl) carbodiimide hydrochloride; 6 h, rt 1.3 Reagents: Phosphorus pentachloride Solvents: Dichloromethane; 0 °C; 18 h, rt 1.4 Reagents: Sodium borohydride Solvents: Methanol; 30 min, 0 °C; 2 min, 0 °C → rt 1.5 Reagents: Water 1.6 Reagents: Sodium borohydride Solvents: Methanol; 30 min, 0 °C; 8 h, 0 °C → rt 1.7 Reagents: Hydrochloric acid Solvents: Methanol, Water; 7 h	Green preparation of (chiral)aporphine alkaloid compounds Assignee: Zhijian Biotechnology (Shanghai) Co., Ltd. China, CN116675642 A 2023-09-01 PatentPak available
---	---

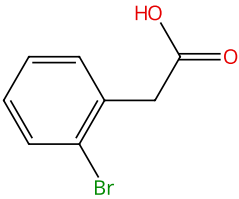


31-313-CAS-21176153 Steps: 1 Yield: 92% 1.1 Reagents: Acetic acid Solvents: Tetrahydrofuran, Water; 40 °C; 1 h, rt; 6 h, 60 °C Experimental Protocols	Improved and large-scale synthesis of N-methyl-D-aspartic acid By: Xi, Liang; et al Current Organic Synthesis (2015), 12(2), 197-201.
---	---

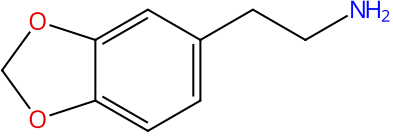


31-614-CAS-37550609 Steps: 1 Yield: 92% 1.1 Reagents: 1-Hydroxybenzotriazole Solvents: Dichloromethane; 30 min, 0 °C 1.2 Reagents: Triethylamine, 1-Ethyl-3-(3'-dimethylaminopropyl) carbodiimide hydrochloride; 6 h, rt 1.3 Reagents: Phosphorus pentachloride Solvents: Dichloromethane; 0 °C; 18 h, rt 1.4 Reagents: Sodium borohydride Solvents: Methanol; 30 min, 0 °C; 2 min, 0 °C → rt 1.5 Reagents: Water 1.6 Reagents: Sodium borohydride Solvents: Methanol; 30 min, 0 °C; 8 h, 0 °C → rt 1.7 Reagents: Hydrochloric acid Solvents: Methanol, Water; 7 h	Green preparation of (chiral)aporphine alkaloid compounds Assignee: Zhijian Biotechnology (Shanghai) Co., Ltd. China, CN116675642 A 2023-09-01 PatentPak available
---	---

Scheme 14 (1 Reaction) Steps: 1 Yield: 92%



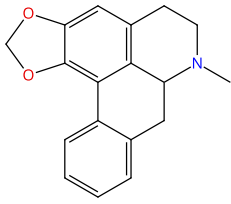
 Suppliers (96)



 Suppliers (39)



 Suppliers (188)

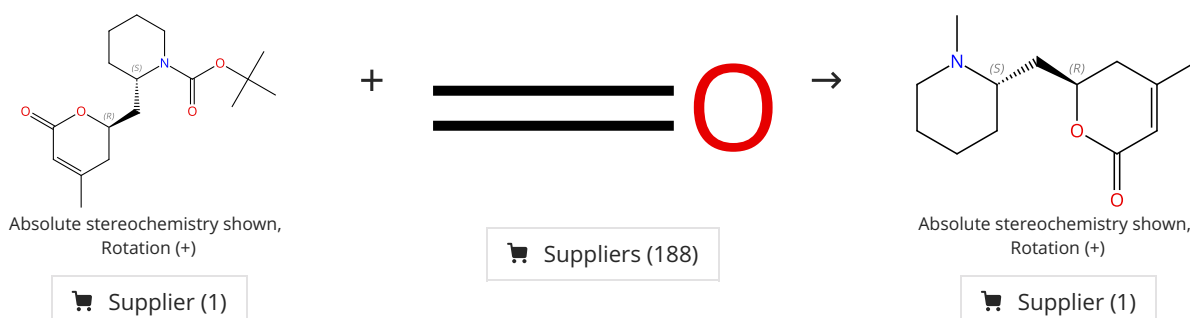


 Suppliers (19)

31-614-CAS-37550610 Steps: 1 Yield: 92% 1.1 Reagents: 1-Hydroxybenzotriazole Solvents: Dichloromethane; 30 min, 0 °C 1.2 Reagents: Triethylamine, 1-Ethyl-3-(3'-dimethylaminopropyl) carbodiimide hydrochloride; 6 h, rt 1.3 Reagents: Phosphorus pentachloride Solvents: Dichloromethane; 0 °C; 18 h, rt 1.4 Reagents: Sodium borohydride Solvents: Methanol; 30 min, 0 °C; 2 min, 0 °C → rt 1.5 Reagents: Water 1.6 Reagents: Sodium borohydride Solvents: Methanol; 30 min, 0 °C; 8 h, 0 °C → rt 1.7 Reagents: Hydrochloric acid Solvents: Methanol, Water; 7 h	Green preparation of (chiral)aporphine alkaloid compounds Assignee: Zhijian Biotechnology (Shanghai) Co., Ltd. China, CN116675642 A 2023-09-01 PatentPak available
---	---

Scheme 15 (1 Reaction)

Steps: 1 Yield: 92%



31-313-CAS-12096939

Steps: 1 Yield: 92%

Synthesis of (+)-Dumetorine and Congeners by Using Flow Chemistry Technologies

1.1 Reagents: Formic acid

Solvents: Acetonitrile; 15 min, 140 °C

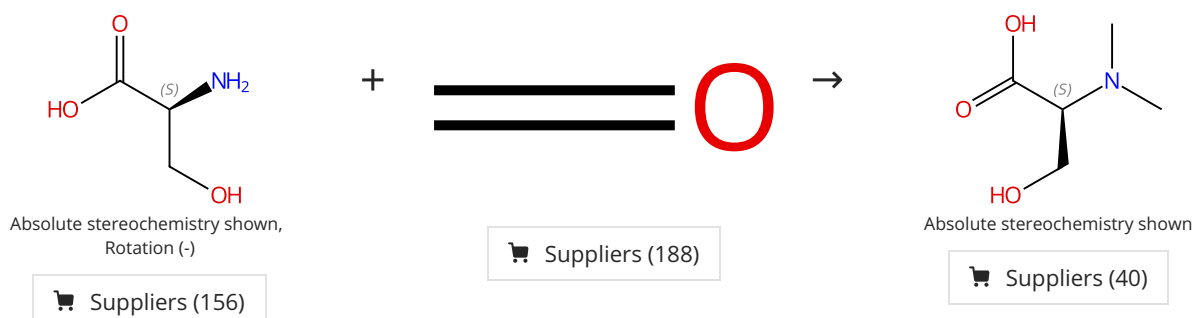
By: Riva, Elena; et al

Experimental Protocols

Chemistry - A European Journal (2011), 17(22), 6221-6226.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 92%



31-313-CAS-4269181

Steps: 1 Yield: 92%

Process for preparation of amino acid N,N-dimethyl derivatives

1.1 0.5 h, rt

1.2 Reagents: Formic acid; 5 h, 100 °C; 0.5 h, 120 °C

Assignee: Northwest Normal University

China, CN101967076 A 2011-02-09

PatentPak available

Scheme 17 (1 Reaction)

Steps: 1 Yield: 92%



31-313-CAS-21113329

Steps: 1 Yield: 92%

Transition metal-free methylation of amines with formaldehyde as the reductant and methyl source

1.1 Reagents: Triethylamine

Solvents: Toluene; 16 h, 130 °C

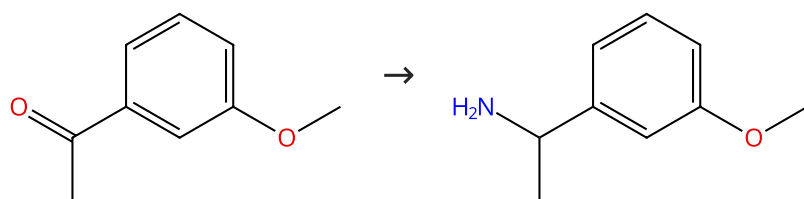
By: Man, Nikki Y. T.; et al

Experimental Protocols

Chimia (2015), 69(6), 345-347.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (90)

Suppliers (70)

31-313-CAS-15240766

Steps: 1 Yield: 92%

Synthesis of *N*-ethyl-*N*-methylcarbamic acid 3-[[1*S*]-1-(dimethylamino)ethyl]phenyl ester, (2*R*,3*R*)-2,3-dihydroxybutanedioate (rivastigmine hydrogen tartrate)

By: Zhou, Lili; et al

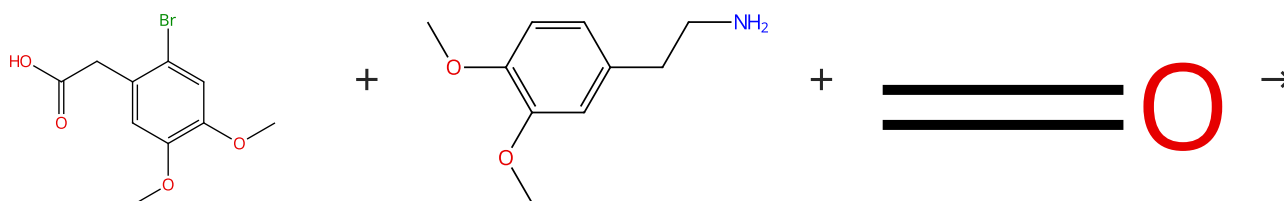
Zhongguo Yiyao Gongye Zazhi (2007), 38(5), 327-329.

1.1 Reagents: Ammonia
Solvents: Methanol; 1 h, rt

1.2 Reagents: Hydrogen
Catalysts: Nickel; 8 h, 610 kPa, 80 °C; 80 °C → rt

Scheme 19 (1 Reaction)

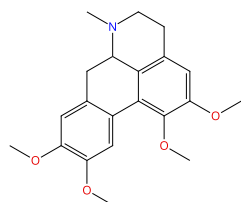
Steps: 1 Yield: 91%



Suppliers (60)

Suppliers (64)

Suppliers (188)



Suppliers (14)

31-614-CAS-37550605

Steps: 1 Yield: 91%

Green preparation of (chiral)aporphine alkaloid compounds

Assignee: Zhijian Biotechnology (Shanghai) Co., Ltd.

China, CN116675642 A 2023-09-01

PatentPak available

1.1 Reagents: 1-Hydroxybenzotriazole
Solvents: Dichloromethane; 30 min, 0 °C

1.2 Reagents: Triethylamine, 1-Ethyl-3-(3'-dimethylaminopropyl) carbodiimide hydrochloride; 6 h, rt

1.3 Reagents: Phosphorus pentachloride
Solvents: Dichloromethane; 0 °C; 18 h, rt

1.4 Reagents: Sodium borohydride
Solvents: Methanol; 30 min, 0 °C; 2 min, 0 °C → rt

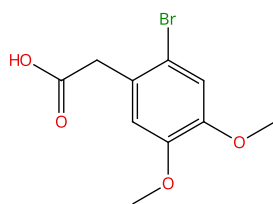
1.5 Reagents: Water

1.6 Reagents: Sodium borohydride
Solvents: Methanol; 30 min, 0 °C; 8 h, 0 °C → rt

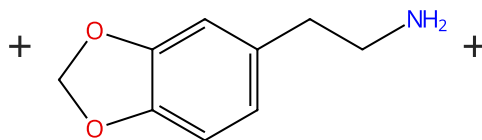
1.7 Reagents: Hydrochloric acid
Solvents: Methanol, Water; 7 h

Scheme 20 (1 Reaction)

Steps: 1 Yield: 90%



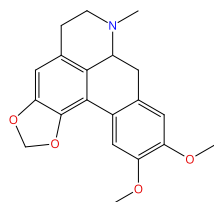
Suppliers (60)



Suppliers (39)



Suppliers (188)



Suppliers (10)

31-614-CAS-37550612

Steps: 1 Yield: 90%

Green preparation of (chiral)aporphine alkaloid compounds

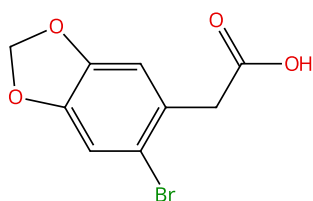
- 1.1 **Reagents:** 1-Hydroxybenzotriazole
Solvents: Dichloromethane; 30 min, 0 °C
- 1.2 **Reagents:** Triethylamine, 1-Ethyl-3-(3'-dimethylaminopropyl) carbodiimide hydrochloride; 6 h, rt
- 1.3 **Reagents:** Phosphorus pentachloride
Solvents: Dichloromethane; 0 °C; 18 h, rt
- 1.4 **Reagents:** Sodium borohydride
Solvents: Methanol; 30 min, 0 °C; 2 min, 0 °C → rt
- 1.5 **Reagents:** Water
- 1.6 **Reagents:** Sodium borohydride
Solvents: Methanol; 30 min, 0 °C; 8 h, 0 °C → rt
- 1.7 **Reagents:** Hydrochloric acid
Solvents: Methanol, Water; 7 h

Assignee: Zhijian Biotechnology (Shanghai) Co., Ltd.
China, CN116675642 A 2023-09-01

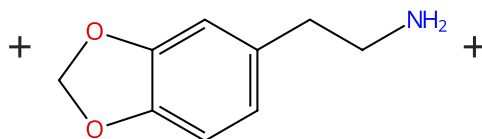
[PatentPak](#) available

Scheme 21 (1 Reaction)

Steps: 1 Yield: 90%



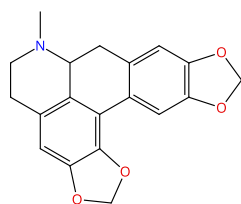
Suppliers (38)



Suppliers (39)

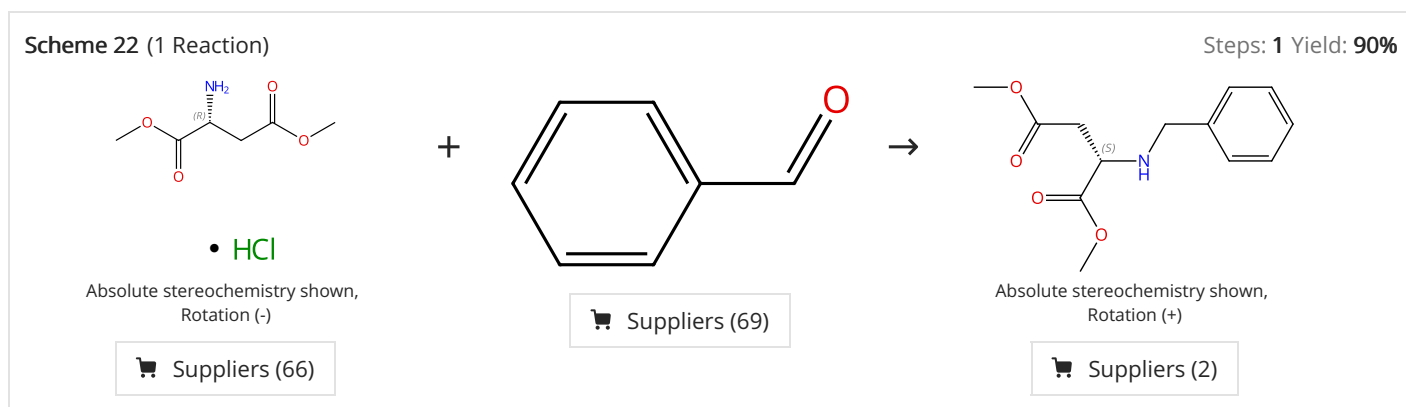


Suppliers (188)

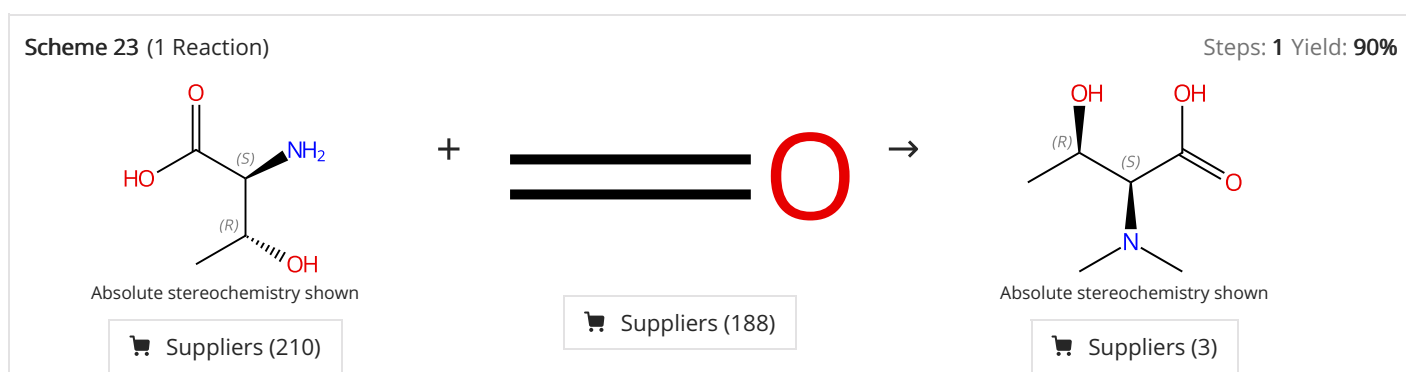


Suppliers (2)

31-614-CAS-37550611 Steps: 1 Yield: 90% 1.1 Reagents: 1-Hydroxybenzotriazole Solvents: Dichloromethane; 30 min, 0 °C 1.2 Reagents: Triethylamine, 1-Ethyl-3-(3'-dimethylaminopropyl) carbodiimide hydrochloride; 6 h, rt 1.3 Reagents: Phosphorus pentachloride Solvents: Dichloromethane; 0 °C; 18 h, rt 1.4 Reagents: Sodium borohydride Solvents: Methanol; 30 min, 0 °C; 2 min, 0 °C → rt 1.5 Reagents: Water 1.6 Reagents: Sodium borohydride Solvents: Methanol; 30 min, 0 °C; 8 h, 0 °C → rt 1.7 Reagents: Hydrochloric acid Solvents: Methanol, Water; 7 h	Green preparation of (chiral)aporphine alkaloid compounds Assignee: Zhijian Biotechnology (Shanghai) Co., Ltd. China, CN116675642 A 2023-09-01 PatentPak available
---	---



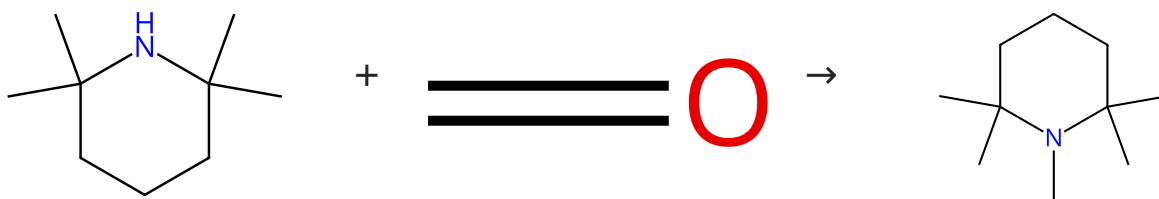
31-313-CAS-15547906 Steps: 1 Yield: 90% 1.1 Reagents: Triethylamine Solvents: Methanol, Water; 0 - 5 °C; 0.5 h, p H 7, 15 °C 1.2 15 °C; 1 h, 15 °C 1.3 Reagents: Sodium borohydride; 5 - 15 °C; 10 h, 60 °C Experimental Protocols	Improved and large-scale synthesis of N-methyl-D-aspartic acid By: Xi, Liang; et al Current Organic Synthesis (2015), 12(2), 197-201.
---	---



31-313-CAS-2144513 Steps: 1 Yield: 90% 1.1 0.5 h, rt 1.2 Reagents: Formic acid; 5 h, 100 °C; 0.5 h, 120 °C	Process for preparation of amino acid N,N-dimethyl derivatives Assignee: Northwest Normal University China, CN101967076 A 2011-02-09 PatentPak available
---	---

Scheme 24 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (92)

Suppliers (188)

Suppliers (72)

31-313-CAS-21112265

Steps: 1 Yield: 89%

Transition metal-free methylation of amines with formaldehyde as the reductant and methyl source

1.1 Reagents: Triethylamine

Solvents: Toluene; 16 h, 130 °C

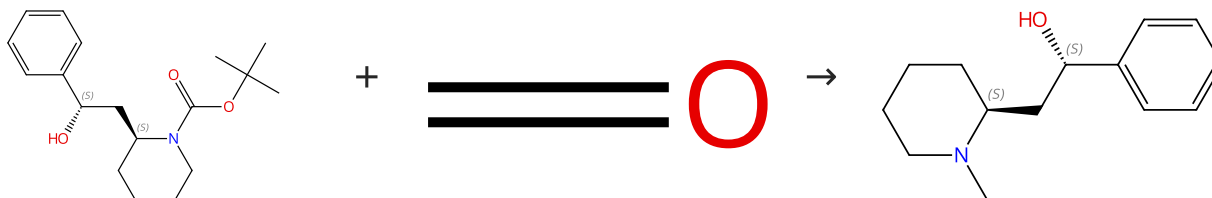
By: Man, Nikki Y. T.; et al

Experimental Protocols

Chimia (2015), 69(6), 345-347.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 89%

Absolute stereochemistry shown,
Rotation (-)

Suppliers (188)

Absolute stereochemistry shown,
Rotation (-)

Suppliers (4)

31-313-CAS-3225451

Steps: 1 Yield: 89%

Synthesis of (+)-Dumetorine and Congeners by Using Flow Chemistry Technologies

1.1 Reagents: Formic acid

Solvents: Acetonitrile; 10 min, 180 °C

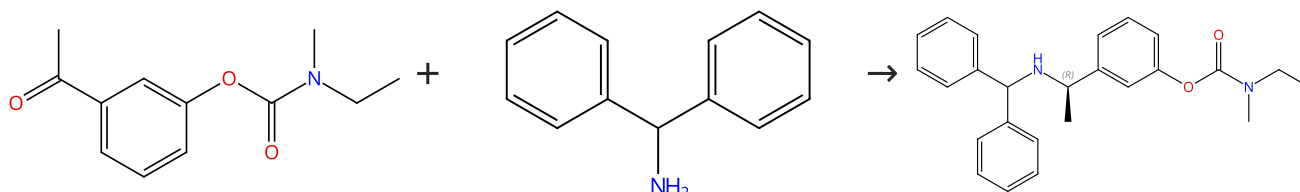
By: Riva, Elena; et al

Experimental Protocols

Chemistry - A European Journal (2011), 17(22), 6221-6226.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 88%



Suppliers (62)

Suppliers (92)

Absolute stereochemistry shown

31-313-CAS-22322814

Steps: 1 Yield: 88%

Synthesis method of rivastigmine optical isomeric intermediate and (R)-rivastigmine for treating Alzheimer's disease

1.1 Reagents: Trifluoroacetic acid, Titanium isopropoxide,

Hydrogen

Catalysts: Iridium

Solvents: Dichloromethane; 20 h, 50 atm, rt → 50 °C

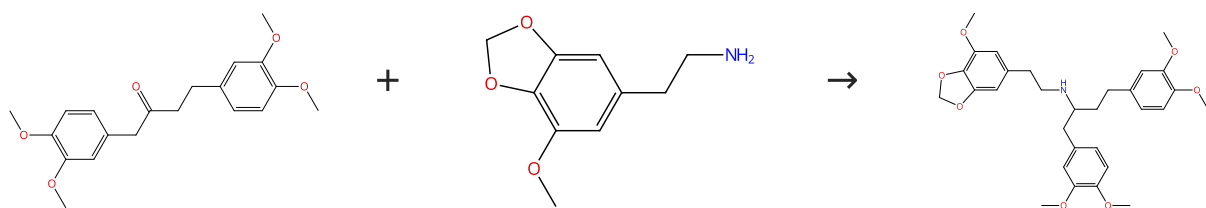
Assignee: Shandong Normal University

China, CN110668976 A 2020-01-10

PatentPak available

Scheme 27 (1 Reaction)

Steps: 1 Yield: 86%



Suppliers (2)

Suppliers (21)

Supplier (1)

Available stereochemistry shown,
Rotation (+)

31-614-CAS-44226799

Steps: 1 Yield: 86%

Total synthesis of natural alkaloids Schwarzinines A-D

1.1 Catalysts: Titanium isopropoxide; 12 h, rt; rt → 0 °C

1.2 Reagents: Sodium borohydride
Solvents: Methanol; 0 °C → rt; 2 h, rt

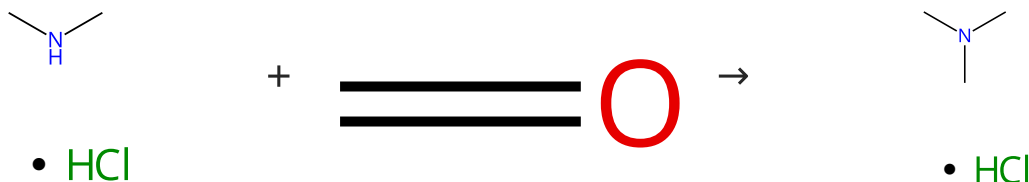
By: Annapurna, K.; et al

Tetrahedron (2024), 155, 133909.

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 84%



Suppliers (67)

Suppliers (188)

Suppliers (78)

31-313-CAS-4174909

Steps: 1 Yield: 84%

A solvent-free and formalin-free Eschweiler-Clarke methylation of amines

1.1 Reagents: Ethanedioic acid hydrate (1:2)

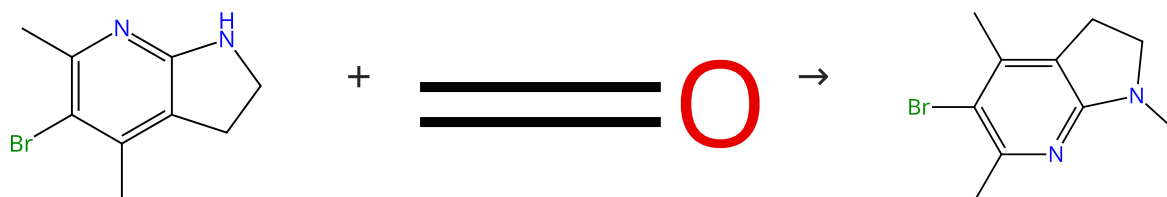
Experimental Protocols

By: Rosenau, Thomas; et al

Synthetic Communications (2002), 32(3), 457-465.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 84%



Suppliers (4)

Suppliers (188)

Suppliers (37)

31-313-CAS-1692812

Steps: 1 Yield: 84%

Synthesis and Reactivity of Some 6- Substituted-2,4-dimethyl-3-pyridinols, a Novel Class of Chain-Breaking Antioxidants

1.1 Reagents: Formic acid
Solvents: Isopropanol; overnight, reflux; cooled1.2 Reagents: Potassium hydroxide
Solvents: Water; basified; 0 °C

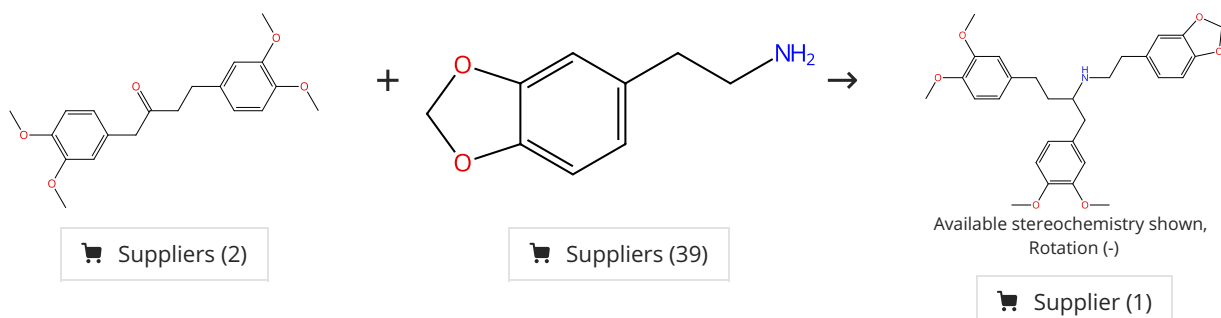
By: Wijtmans, Maikel; et al

Journal of Organic Chemistry (2004), 69(26), 9215-9223.

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 84%



31-614-CAS-44226792

Steps: 1 Yield: 84%

Total synthesis of natural alkaloids Schwarzinines A-D

1.1 Catalysts: Titanium isopropoxide; 12 h, rt; rt → 0 °C

1.2 Reagents: Sodium borohydride
Solvents: Methanol; 0 °C → rt; 2 h, rt

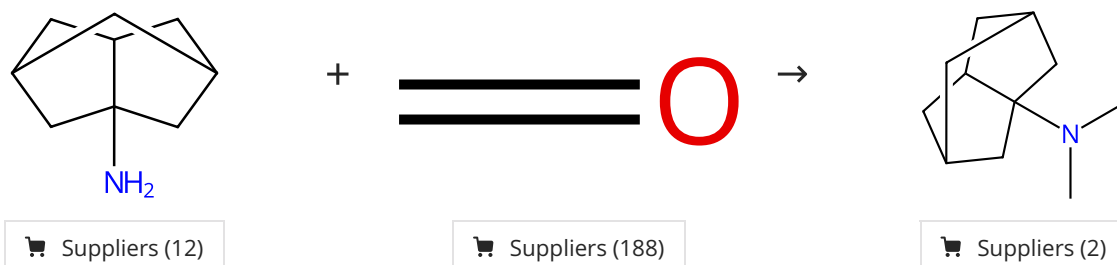
By: Annapurna, K.; et al

Tetrahedron (2024), 155, 133909.

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 84%



31-313-CAS-5340665

Steps: 1 Yield: 84%

Synthesis of N,N-dimethyl-3-noradamantanamine N-oxide

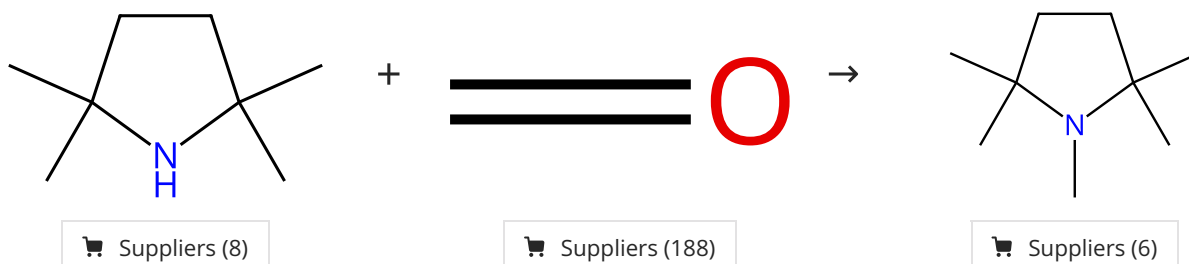
No Data Available

By: Casanova, Joseph; et al

Synthetic Communications (1993), 23(2), 245-51.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 83%



31-313-CAS-4603778

Steps: 1 Yield: 83%

Ring-opening reactions. VII. Ring-size effects on the reactivities of 1,1,2,2,ω,ω-hexamethyl cyclic ammonium ions in the exocyclic elimination reaction

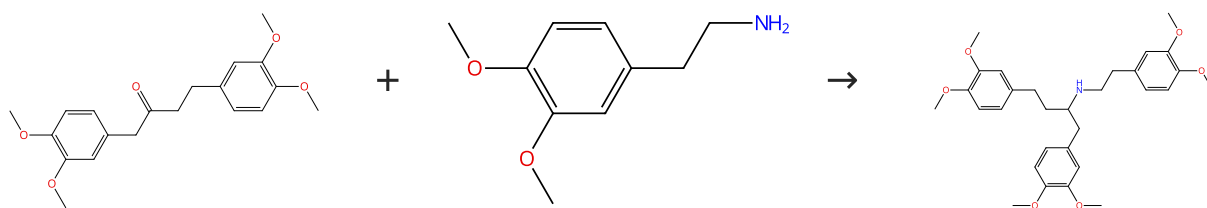
1.1 Reagents: Formic acid
Solvents: Diethyl ether, Water

By: Frachey, Giuseppe; et al

Gazzetta Chimica Italiana (1993), 123(8), 463-6.

Scheme 33 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (2)

Suppliers (64)

Supplier (1)

31-614-CAS-44226797

Steps: 1 Yield: 82%

Total synthesis of natural alkaloids Schwarzinicines A-D

1.1 Catalysts: Titanium isopropoxide; 12 h, rt; rt → 0 °C

By: Annapurna, K.; et al

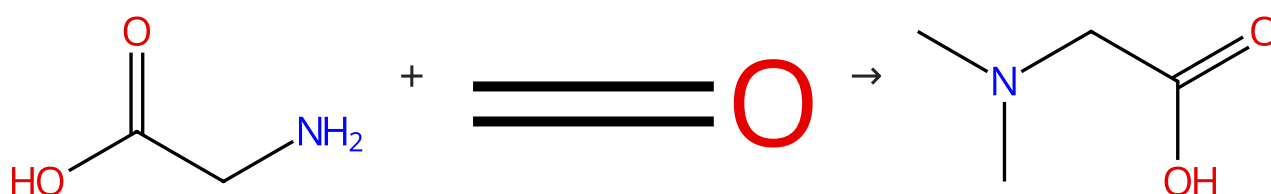
1.2 Reagents: Sodium borohydride
Solvents: Methanol; 0 °C → rt; 2 h, rt

Tetrahedron (2024), 155, 133909.

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (211)

Suppliers (188)

Suppliers (94)

31-313-CAS-6152902

Steps: 1 Yield: 82%

An anomalous Eschweiler-Clarke reaction

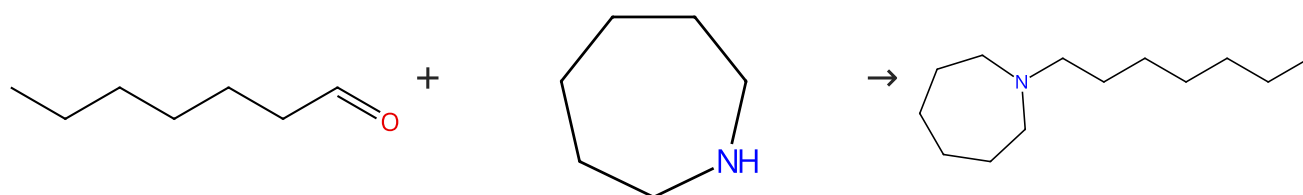
1.1 Reagents: Formic acid
Solvents: Formic acid

By: Rahal, Said; et al

Tetrahedron Letters (1991), 32(31), 3847-8.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 81%



Suppliers (65)

Suppliers (50)

Suppliers (2)

31-313-CAS-15232607

Steps: 1 Yield: 81%

N-alkylation of amines under microwave irradiation: modified Eschweiler-Clarke reaction

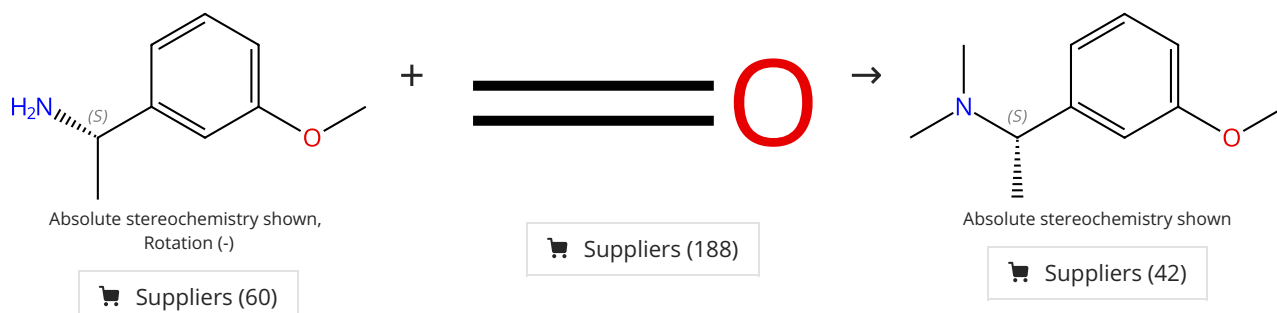
1.1 Reagents: Formic acid

By: Torchy, Severine; et al

Journal of Chemical Research, Synopses (2001), (7), 292-293.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 81%



31-313-CAS-4692978

Steps: 1 Yield: 81%

Synthesis of *N*-ethyl-*N*-methylcarbamic acid 3-[[*(1S)*-1-(dimethylamino)ethyl]phenyl ester, (*2R,3R*)-2,3-dihydroxybutanedioate (rivastigmine hydrogen tartrate)

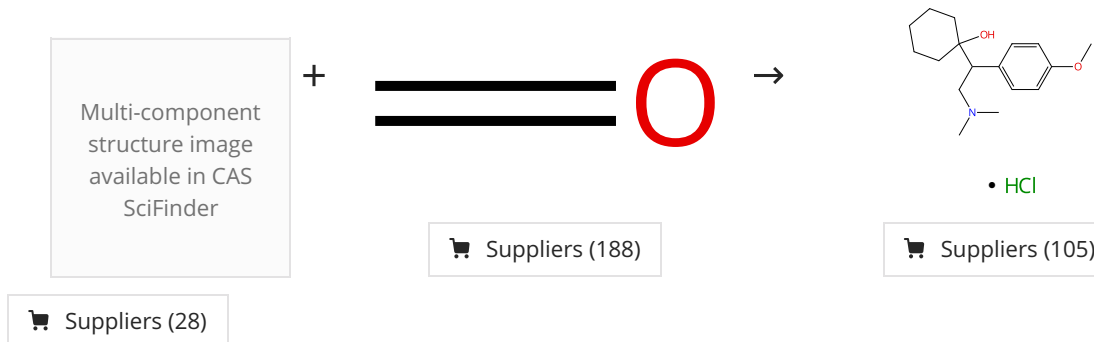
By: Zhou, Lili; et al

Zhongguo Yiyao Gongye Zazhi (2007), 38(5), 327-329.

- 1.1 Reagents: Formic acid
Solvents: Water; 10 h, 95 - 100 °C

Scheme 37 (1 Reaction)

Steps: 1 Yield: 80%



31-313-CAS-12477097

Steps: 1 Yield: 80%

A process for the preparation of venlafaxine hydrochloride

Assignee: Hikal Limited

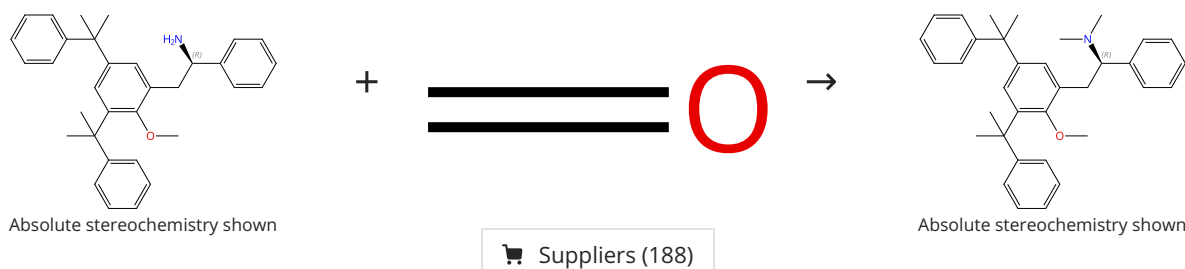
India, IN2006CH00257 A 2008-11-28

[PatentPak](#) available

- 1.1 Reagents: Hydrogen
Catalysts: Nickel
Solvents: Methanol; 4 kg/cm², rt; 8 h, 20 - 25 °C
- 1.2 Reagents: Potassium carbonate
Solvents: Water; pH 11
- 1.3 Reagents: Hydrochloric acid
Solvents: Isopropanol, Ethyl acetate; 2 - 3 h, pH 2, 20 - 25 °C; 1 - 2 h, reflux

Scheme 38 (1 Reaction)

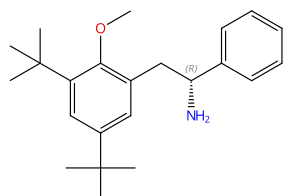
Steps: 1 Yield: 79%



31-313-CAS-22106624 Steps: 1 Yield: 79% 1.1 Reagents: Formic acid Solvents: Water; 12 h, reflux 1.2 Reagents: Sodium bicarbonate Solvents: Water; pH 9 - 10	Chiral 1,4-aminoalkylphenols for enantioselective diethylzinc addition to aldehydes By: Dilek, Oemer; et al Turkish Journal of Chemistry (2019), 43(2), 612-623.
---	--

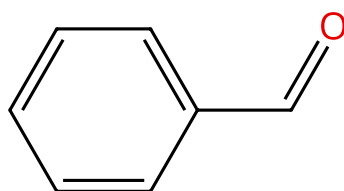
Scheme 39 (1 Reaction)

Steps: 1 Yield: 79%

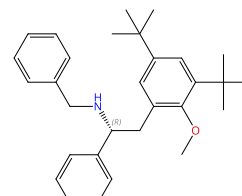


Absolute stereochemistry shown,
Rotation (-)

+



→



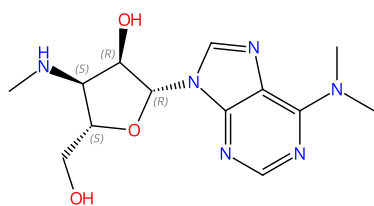
Absolute stereochemistry shown

Suppliers (69)

31-313-CAS-22163931 Steps: 1 Yield: 79% 1.1 Reagents: Sulfuric acid magnesium salt (1:1) Solvents: Dichloromethane; 16 h, rt 1.2 Reagents: Sodium cyanoborohydride Solvents: Acetonitrile; 15 min, 0 °C 1.3 Reagents: Acetic acid; 18 h, rt	Chiral 1,4-aminoalkylphenols for enantioselective diethylzinc addition to aldehydes By: Dilek, Oemer; et al Turkish Journal of Chemistry (2019), 43(2), 612-623.
--	--

Scheme 40 (1 Reaction)

Steps: 1 Yield: 78%

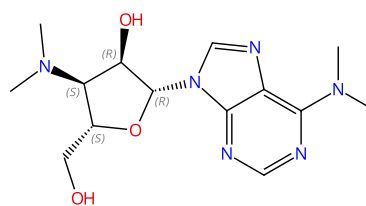


Absolute stereochemistry shown

+



→



Absolute stereochemistry shown

Supplier (1)

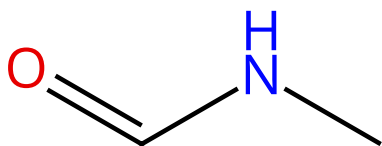
Suppliers (188)

Suppliers (2)

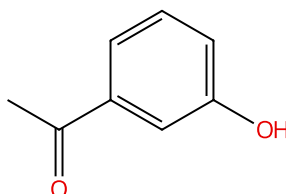
31-313-CAS-11114062 Steps: 1 Yield: 78% 1.1 Reagents: Formic acid Solvents: Water	Nucleosides from 3-deoxy-3-methylamino-D-ribofuranose By: Lee, William Wei; et al Journal of Organic Chemistry (1970), 35(11), 3808-14.
--	---

Scheme 41 (1 Reaction)

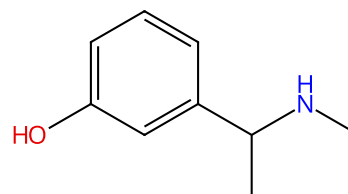
Steps: 1 Yield: 78%



+



→



Suppliers (74)

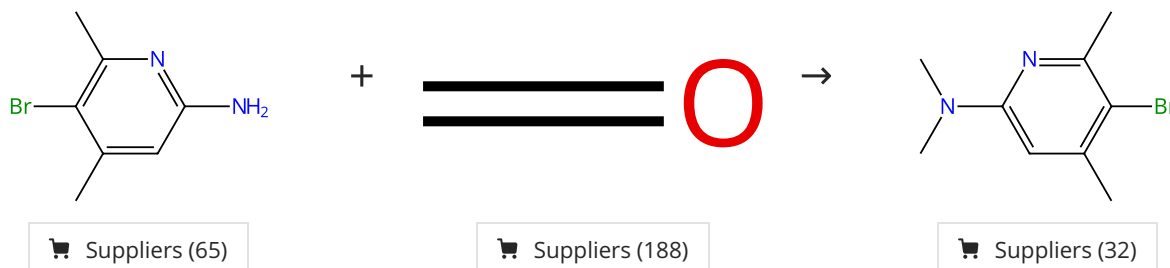
Suppliers (104)

Suppliers (21)

31-313-CAS-18643368 Steps: 1 Yield: 78% 1.1 Reagents: Formic acid; rt → 110 °C; 8 h, 110 °C; 110 °C → 70 °C 1.2 Reagents: Sodium hydroxide Solvents: Water; 2 h, reflux; reflux → rt 1.3 Reagents: Hydrochloric acid Solvents: Water; pH 8 - 9	Preparation of 3-[1-(dimethylamino)ethyl]phenol as intermediate of rivastigmine for treating central nervous system diseases Assignee: Zhejiang Jingxin Pharmaceutical Co., Ltd. China, CN104987294 A 2015-10-21 PatentPak available
--	--

Scheme 42 (1 Reaction)

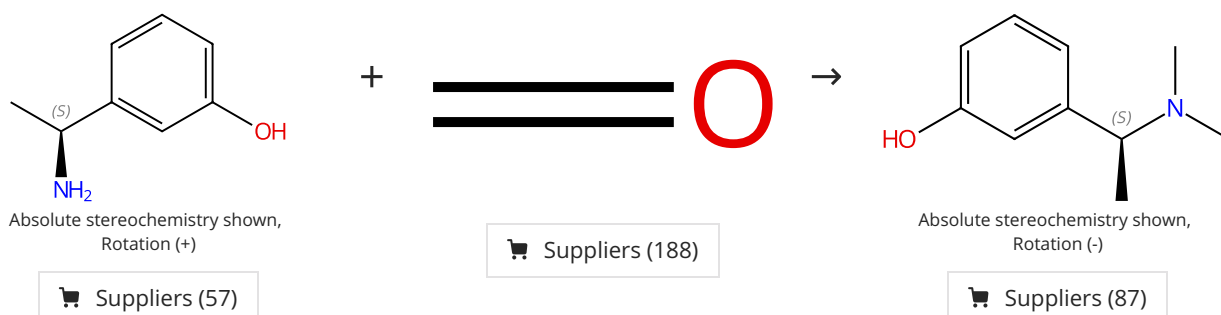
Steps: 1 Yield: 78%



31-313-CAS-8748837 Steps: 1 Yield: 78% 1.1 Solvents: Formic acid; overnight, reflux; cooled 1.2 Reagents: Sodium hydroxide Solvents: Dichloromethane, Water; rt Experimental Protocols	Synthesis and Reactivity of Some 6- Substituted-2,4-dimethyl-3-pyridinols, a Novel Class of Chain-Breaking Antioxidants By: Wijtmans, Maikel; et al Journal of Organic Chemistry (2004), 69(26), 9215-9223.
--	---

Scheme 43 (1 Reaction)

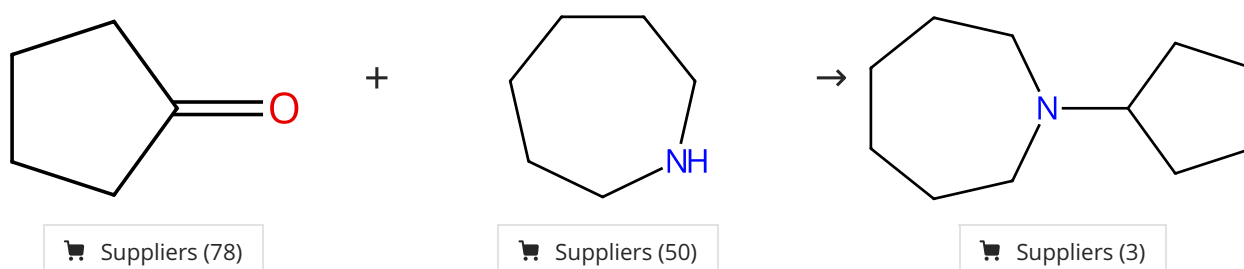
Steps: 1 Yield: 75%



31-313-CAS-18233263 Steps: 1 Yield: 75% 1.1 Reagents: Formic acid Solvents: Water; 2 h, 25 °C; 25 °C → 80 °C; 20 h, 80 °C 1.2 Reagents: Hydrochloric acid Solvents: Water 1.3 Reagents: Ammonia Solvents: Water; pH 9	Process for the asymmetric synthesis of (S)-rivastigmine Assignee: Biortus Biosciences Co., Ltd. China, CN103319374 A 2013-09-25 PatentPak available
---	--

Scheme 44 (1 Reaction)

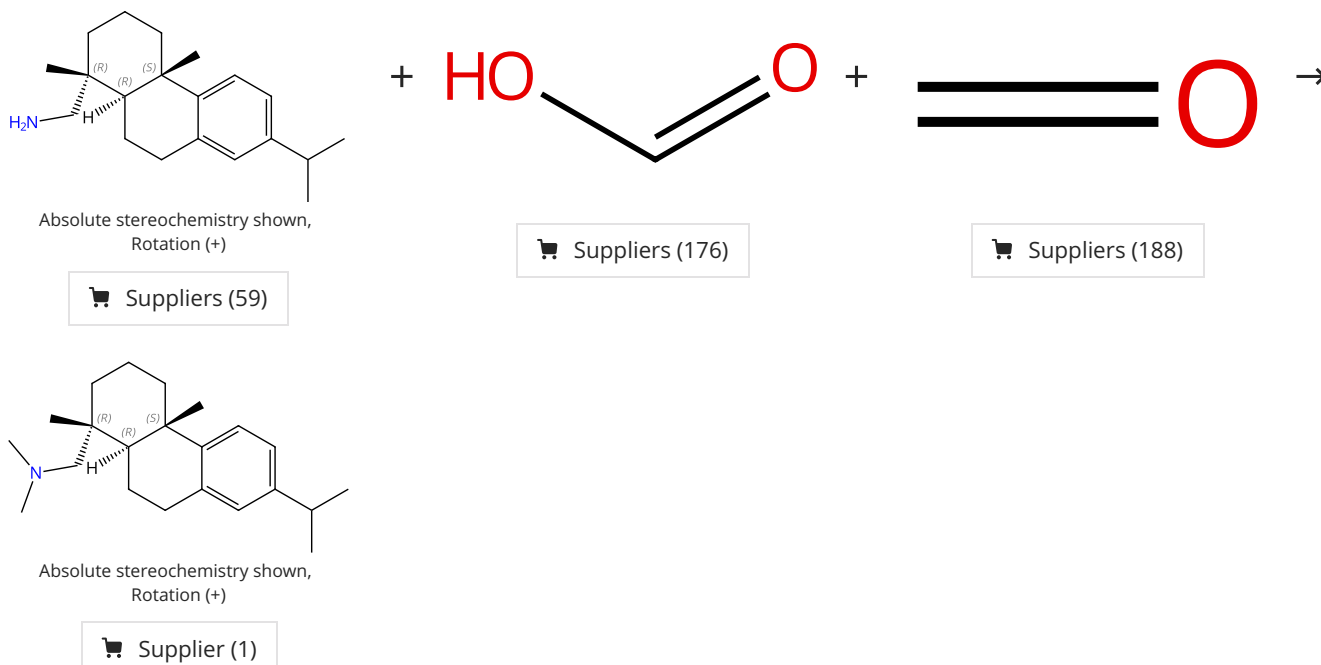
Steps: 1 Yield: 75%



31-313-CAS-12819906	Steps: 1 Yield: 75% N-alkylation of amines under microwave irradiation: modified Eschweiler-Clarke reaction
1.1 Reagents: Formic acid	By: Torchy, Severine; et al Journal of Chemical Research, Synopses (2001), (7), 292-293.

Scheme 45 (1 Reaction)

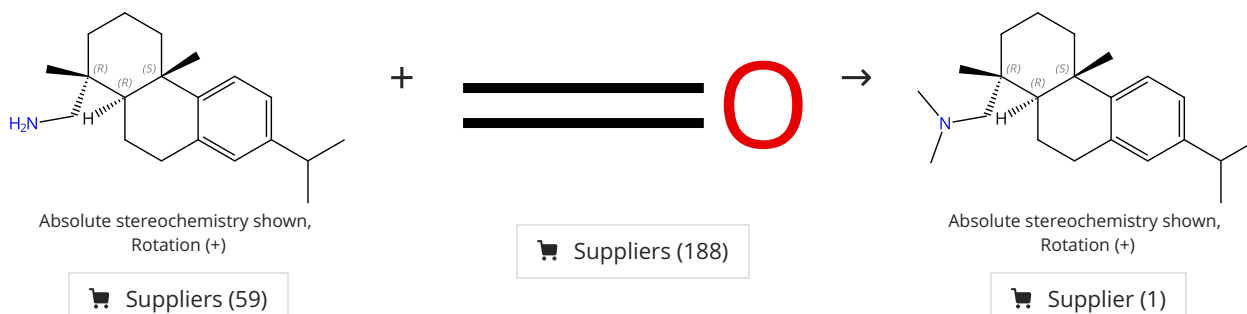
Steps: 1 Yield: 73%



31-313-CAS-21095078	Steps: 1 Yield: 73% Preparation of 3-chloro-2-hydroxypropyl dimethyl dehydro abietyl ammonium chloride and its surface activities
1.1 Solvents: Ethanol, Water; 1 h, rt	By: Zhu, Xue-mei; et al Linchan Huaxue Yu Gongye (2014), 34(3), 116-120.

Scheme 46 (2 Reactions)

Steps: 1 Yield: 73%

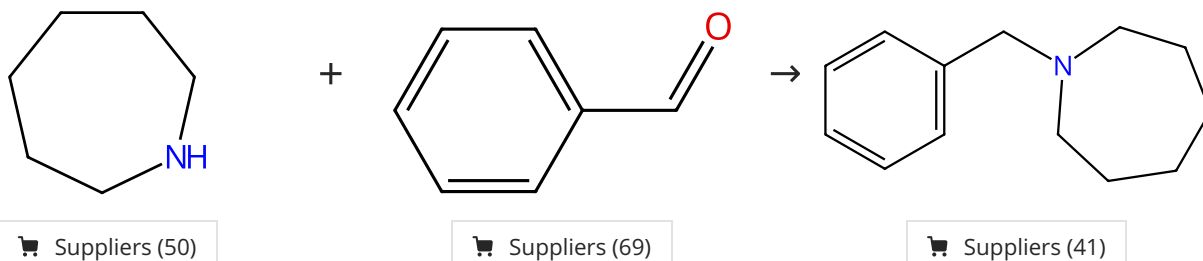


31-313-CAS-21202222	Steps: 1 Yield: 73% Synthesis of allyl dimethyl dehydroabietyl ammonium chloride and its surface activities
1.1 Reagents: Formic acid Solvents: Ethanol, Water; rt; 1 h, rt; rt → 87 °C; 12 h, 87 °C	By: Xu, Da-Xiong; et al
Experimental Protocols	Journal of the Chemical Society of Pakistan (2015), 37(1), 105-110.

31-313-CAS-18066903	Steps: 1 Yield: 73%	3-Chloro-2-hydroxypropyl Dimethyl Dehydroabietyl Ammonium Chloride: Synthesis, Characterization, and Physicochemical Properties
1.1 Reagents: Formic acid Solvents: Ethanol, Water; rt; 1 h, rt; rt → 87 °C; 12 h, 87 °C		By: Pei, Li-jun; et al
1.2 Reagents: Sodium hydroxide, Sodium chloride Solvents: Water		Journal of Surfactants and Detergents (2014), 17(3), 493-499.
Experimental Protocols		

Scheme 47 (1 Reaction)

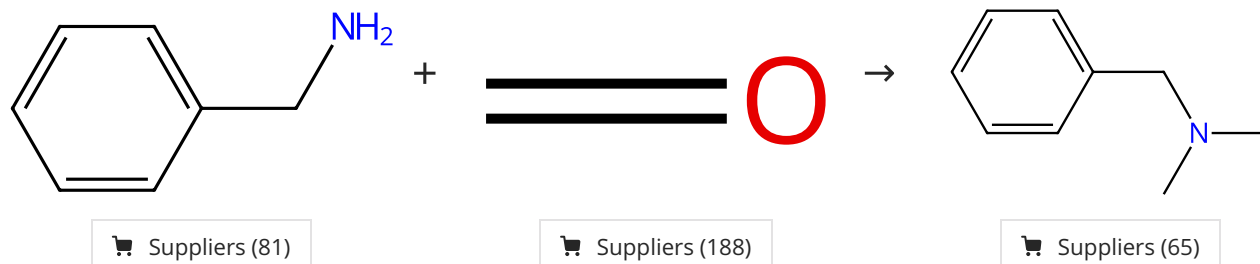
Steps: 1 Yield: 66%



31-313-CAS-4279726	Steps: 1 Yield: 66%	N-alkylation of amines under microwave irradiation: modified Eschweiler-Clarke reaction
1.1 Reagents: Formic acid		By: Torchy, Severine; et al
		Journal of Chemical Research, Synopses (2001), (7), 292-293.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 66%



31-313-CAS-6111007	Steps: 1 Yield: 66%	N-alkylation of amines under microwave irradiation: modified Eschweiler-Clarke reaction
1.1 Reagents: Formic acid		By: Torchy, Severine; et al
		Journal of Chemical Research, Synopses (2001), (7), 292-293.